Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

arr[i-1]

arr 0 = arr 0 % 4
arr 1 = 1 % 4
2 % 4
3 % 4

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=4;i++)
4       (i-1)[arr] = (i%4);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 2;
8   }
9  int main(){
10    int arr[5] = {1,1,1,1,1};
11    func1(arr+1);
12    for(int i=0; i<5; i++)
13          printf("%d ", i[arr] );
14    printf("\n");
15    func2(arr, *arr);
16    for(int i=0; i<5; i++)
17          printf("%d ", i[arr] );
18    return 0;
19 }
```

[1, 1, 1, 1]

0 2 3

Output
1 0 1 2 3
3 0 1 2 3

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
 int temp = arr + 2;
 a = &temp;
}
```

Explain your answer

Yes, because now a is pointing to the address of temp, and will store temp's address in arr[0]

Q. 2: Write the output of the following sequence of instructions **long arr2[60]; int arr1[50], a1=(&arr2[40]), a2=(&arr2[30]), b1=(&arr1[40]), b2=(&arr1[30]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

80 40

a1 - a2 is difference between 10 consecutive memory spaces, each containing 8 bytes of memory. Hence $a_1 - a_2 = 10 \times 8$
Similarly for $b_1 - b_2$